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Inventor(s)	Lee; Dong Hee et al.

Motor stator having a stator core with a wound sheet cylinder structure

Abstract

A stator of a motor has a structure including a plate portion coupled to a support structure for fixing a stator core and protrusions inserted into a bobbin. The support structure for fixing the stator core has a structure into which the protrusions of the stator core are inserted and extend into and on which the plate portion of the stator core is inserted and seated. The plate portion of the stator core is joined to the support structure while being coplanar with the support structure. Accordingly, the distance of a gap between the stator core of the stator and a permanent magnet of a rotor is minimized, thereby improving the performance of the motor.

Inventors: **Lee; Dong Hee (Hwaseong-si, KR), Choi; Jae Wan (Anyang-si, KR), Byun; Sung Gon (Hwaseong-si, KR), Kim; Ji Yeon (Yongin-si, KR)**

Applicant: **HYUNDAI MOTOR COMPANY** (Seoul, KR); **KIA CORPORATION** (Seoul, KR)

Family ID: **88663307**

Assignee: **HYUNDAI MOTOR COMPANY (Seoul, KR); KIA CORPORATION (Seoul, KR)**

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Primary Examiner: Singh; Alexander A

Assistant Examiner: Mann, Jr.; Charlie Frank

Attorney, Agent or Firm: Lempia Summerfield Katz LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims under 35 U.S.C. § 119(a) the benefit of priority to Korean Patent Application No. 10-2022-0058803, filed on May 13, 2022, the entire contents of which are incorporated herein by reference.
BACKGROUND
(a) Technical Field
(2) The present disclosure relates to a motor stator, and more particularly, to a motor stator having a configuration by which the distance of a gap (hereinafter, referred to as the “gap distance”) between the stator and a rotor of a motor may be minimized, thereby improving the performance of the motor.
(b) Background Art
(3) In general, eco-friendly vehicles, such as an electric vehicle, a hybrid electric vehicle, and a fuel cell electric vehicle, are respectively provided with a motor based on an axial flux permanent magnet (AFPM) as a motive power source.
(4) A stator of such a motor may include a stator core in which a plurality of steel sheets are stacked on each other, a support structure formed of a polymer to fix a stator core, a coil wound on the stator core, and the like.
(5) Rotors of the motor are respectively fabricated as a structure obtained by attaching a plurality of permanent magnets to one surface of the rotor core and are disposed on both sides of the stator.
(6) The performance of the motor may be determined depending on the gap distance between the stator core of the stator and each of the permanent magnets of the rotors.
(7) For example, with increases in the gap distance between the stator core of the stator and the permanent magnet of the rotor, there may be degradations in performance of the motor, such as a decrease in motor torque or counter-electromotive force. In contrast, with decreases in the gap distance between the stator core of the stator and the permanent magnet of the rotor, there may be improvements in performance of the motor, such as increases in motor torque or counter-electromotive force.
(8) However, according to the related art, there may be a problem in that the gap distance between the stator core of the stator and the permanent magnet of the rotor may be increased due to the

thickness of the support structure for fixing the stator core, thereby degrading the performance of the motor.

SUMMARY

(9) Various embodiments are directed to a stator of a motor in which a stator core is provided with a structure comprised of a plate portion coupled to a support structure for fixing the stator core and protrusions inserted into a bobbin. The support structure for fixing the stator core has a structure into which the protrusions of the stator core are extendedly inserted and into and on which the plate portion of the stator core is inserted and seated. Thus, the plate portion of the stator core is joined to the support structure while being coplanar with the support structure. Accordingly, it is possible to minimize the gap distance between the stator core of the stator and a permanent magnet of a rotor, thereby improving the performance of the motor.

(10) In an embodiment, a motor stator includes a pair of stator cores each having a structure including a plate portion and protrusions provided on one surface of the plate portion. The motor stator also includes stator core-fixing support structures each including through-holes into which the protrusions are inserted to extend and a recess portion into and on which the plate portion is inserted and seated. The motor stator further includes a bobbin having insertion holes therein into which the protrusions, which have passed through the through-holes of the stator core-fixing support structures, are inserted. A coil is wound on an outer surface of the bobbin. The motor stator also includes a housing mounted on inner circumferential portions and outer circumferential portions of the support structures.

(11) According to an implementation, each of the stator cores may be fabricated by forming a cylindrical structure by winding a steel sheet, which may have a predetermined width, a plurality of times and then machining a plurality of cut recess portions on one surface of the cylindrical structure such that the cut recess portions are equidistantly spaced apart from each other in the circumferential direction. Thus, the bottom surfaces of the cut recess portions form the plate portion and the plurality of protrusions are formed to alternate with the cut recess portions.

(12) Each of the support structures may be configured such that an inner ring and an outer ring are integrally connected by a plurality of connecting bars equidistantly disposed in the circumferential direction.

(13) The support structures may be fabricated from one material of polyether ether ketone (PEEK) or polyphthalamide (PPA) among engineered plastic materials by insert injection so as to be joined to the stator cores, respectively.

(14) The through-holes of the support structures may be formed to alternate with the connecting bars of the support structures. The recess portion, into and on which the plate portion of a corresponding one of the stator cores is inserted and seated, may be provided on outer surface portions of the connecting bars connecting the inner ring and the outer ring.

(15) The inner ring and the outer ring may have the same thickness. The thickness of each of the connecting bars may be lower than the thickness of either the inner ring or the outer ring. Also, the recess portion may be formed on the outer surface portions of the connecting bars connecting an outer diameter surface of the inner ring and an inner diameter surface of the outer ring.

(16) The depth of each of the recess portions of the support structures may be the same as the thickness of each of the plate portions of the stator cores.

(17) Thus, when the plate portion of each of the stator cores is inserted into and seated on the recess portion of a corresponding one of the support structures, an outer surface of the plate portion of the stator core and the outer surfaces of the inner ring and the outer ring of the support structure may be disposed coplanar.

(18) The housing may include an inner housing joined to inner diameter surfaces of the support structures by laser welding or thermal fusion. The housing may further include an outer housing joined to outer diameter surfaces of the support structures by laser welding or thermal fusion in order to seal the bobbin on which the coil is wound.

(19) The outer housing may include a cooling oil inlet and a cooling oil outlet configured to cool the coil wound on the bobbin by immersing the coil in cooling oil.
(20) Each of the stator cores may be fabricated as a structure obtained by stacking and joining a plurality of divided steel pieces. A protrusion and a plate portion may be integrated in each steel piece. The protrusions may be configured to extend through the through-holes of a corresponding one of the support structures to be inserted into the insertion holes of the bobbin. The plate portions may be configured to be inserted into and seated on the recess portion of a corresponding one of the support structures.
(21) The protrusion of each of the divided steel pieces of the stator cores may have a coupling recess or a coupling protrusion on a distal end thereof.
(22) Thus, when the protrusions, each having the coupling recess of the divided steel pieces, are inserted into the insertion holes of the bobbin, respectively, from one side of the bobbin, and the protrusions are inserted into the insertion holes of the bobbin, respectively, from the other side of the bobbin, the coupling protrusions may be inserted into and fastened to the coupling recesses, respectively.
(23) Through the above configuration, the present disclosure provides the following effects.
(24) First, the plate portion of each of the stator cores may be joined to the corresponding stator core-fixing support structure while being coplanar with the support structure. Accordingly, it is possible to minimize the gap distances between the stator core of the stator and the permanent magnets of the rotor, thereby improving the performance of the motor. For example, motor torque, counter-electromotive force, or the like may be increased.
(25) Second, due to the configuration by which the bobbin and the coil are sealed inside the housing and the stator cores, the coil may be directly cooled using the cooling oil. For example, the cooling oil may be injected through the cooling oil inlet formed in the outer housing, the coil wound on the bobbin may be cooled by immersing the coil in the cooling oil, the cooling oil may be discharged through the cooling oil outlet formed in the outer housing, and the like.
(26) Third, the stator core-fixing support structures may be formed of a non-magnetic engineered plastic material, thereby preventing an electromagnetic loss, such as an eddy current loss.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above and other features of the present disclosure are described in detail with reference to certain examples thereof illustrated in the accompanying drawings which are given herein below by way of illustration only, and thus are not limitative of the present disclosure, and wherein:
(2) FIG. **1** is an exploded perspective view illustrating a motor including a stator according to the present disclosure;
(3) FIG. **2** is a perspective view illustrating the stator core of a stator according to an embodiment of the present disclosure;
(4) FIG. **3** is a perspective view illustrating the support structure of the stator according to the present disclosure;
(5) FIG. **4** is a perspective view illustrating the coupling state between the stator core and the support structure of the stator according to the present disclosure;
(6) FIG. **5** is a side view illustrating the coupling state between the stator core and the support structure of the stator according to the present disclosure;
(7) FIG. **6** is a perspective view illustrating a state in which the stator cores coupled to the support structures of the stator are coupled to the bobbin according to the present disclosure;
(8) FIG. **7** is a perspective view illustrating a state in which the bobbin, to which the stator cores coupled to the support structures of the stator are coupled, and the housing are coupled according to

the present disclosure;
(9) FIG. **8** is a perspective view illustrating an assembled state of the stator according to the present disclosure;
(10) FIG. **9** is a perspective view illustrating a state in which the rotor is disposed adjacent to the stator according to the present disclosure;
(11) FIG. **10** is a cross-sectional view illustrating, in a comparative manner, a decrease in the gap distance between the stator core and the permanent magnet of the rotor according to the present disclosure;
(12) FIG. **11** is a graph illustrating changes in the performance of the motor depending on the gap distance between the stator core and the permanent magnet of the stator according to the present disclosure;
(13) FIG. **12** is a perspective view illustrating stator cores of a stator according to another embodiment of the present disclosure;
(14) FIG. **13** is a perspective view illustrating the coupling state between the stator cores and support structures according to another embodiment of the present disclosure; and
(15) FIG. **14** is a cross-sectional view illustrating a main-part cross-sectional view illustrating a state in which a protrusion of the stator core is inserted into and engaged with the inside of the bobbin according to another embodiment of the present disclosure.
DETAILED DESCRIPTION
(16) Hereinafter, embodiments of the present disclosure are described in detail with reference to the accompanying drawings.
(17) FIG. **1** in the accompanying drawings is an exploded perspective view illustrating a motor including a stator **100** according to the present disclosure.
(18) As illustrated in FIG. **1**, the stator **100** according to the present disclosure includes: a pair of stator cores **110**; stator core-fixing support structures (e.g., support structures) **120** respectively coupled to a corresponding one of the stator cores **110**; a bobbin **130** on which a coil **132** is wound; a housing **140**; and the like.
(19) Each of the stator cores **110** is fabricated with a structure in which protrusions **112** protrude from one surface of a plate portion **111**.
(20) According to an embodiment of the present disclosure, as illustrated in FIG. **2**, the stator cores **110** may be fabricated by forming a cylindrical structure **113** with a steel sheet, which has a predetermined width, wound a plurality of times around the cylinder structure **113** in the manner of winding a windup spring. The stator cores **110** may then be machined to include a plurality of cut recess portions **114** on one surface of the cylindrical structure **113** by a wire cutting method such that the cut recess portions **114** are equidistantly spaced apart from each other in the circumferential direction.
(21) Consequently, the stator cores **110** respectively including the plurality of cut recess portions **114**, the bottom surfaces of which are formed as the plate portion **111**, and the plurality of protrusions **112** alternating with the cut recess portions **114**, may be realized.
(22) As illustrated in FIG. **3**, each of the support structures **120** may be fabricated with a structure in which through-holes **122** and a recess portion **121** are formed. The protrusions **112** of the corresponding stator core **110** are inserted into and extend through the through-holes **122** and the plate portion **111** is inserted into and seated on the recess portion **121**.
(23) More specifically, each of the support structures **120** is configured such that an inner ring **123** and an outer ring **125** are integrally connected by a plurality of connecting bars **124** equidistantly disposed in the circumferential direction. Thus, the through-holes **122** may be formed to alternate with the connecting bars **124** of the corresponding support structure **120**. The protrusions **112** are inserted and extend into the through-holes **122**. The recess portion **121** may be provided on outer surface portions of the connecting bars **124** connecting the inner ring **123** and the outer ring **125**. The plate portion **111** of the corresponding stator core **110** is inserted into and seated on the recess

portion **121**.
(24) The inner ring **123** and the outer ring **125** of the support structure **120** have the same thickness and the thickness of each of the connecting bars **124** is lower than the thickness of either the inner ring **123** or the outer ring **125**. Thus, the recess portion **121** may be easily formed on the outer surface portions of the connecting bars **124** connecting the outer diameter surface of the inner ring **123** and the inner diameter surface of the outer ring **125**.
(25) Particularly, the depth of the recess portion **121** of the support structure **120** is formed to be the same as the thickness of the plate portion **111** of the stator core **110**.
(26) Thus, in order to couple the stator core **110** and the support structure **120**, when the protrusions **112** of the stator core **110** are inserted and extend into the through-holes **122** of the support structure **120** and the plate portion **111** of the stator core **110** is inserted into and seated on the recess portion **121** of the support structure **120**, the outer surface of the plate portion **111** of the stator core **110** and the outer surfaces of the inner ring **123** and the outer ring **125** of the support structure **120** may be disposed coplanar as illustrated in FIGS. **4** and **5**.
(27) Particularly, the support structure **120** may be fabricated from a material such as polyether ether ketone (PEEK) or polyphthalamide (PPA) having superior strength and less deformation among engineered plastic materials. The support structure **120** may be fabricated by insert injection so as to be joined to the stator core **110**, respectively.
(28) For example, the support structure **120**, which has the above-described structure, may be integrally joined to the stator core **110** by insert injection via inserting the stator core **110** into a cavity of an injection mold (not shown) and filling the cavity of the injection mold with the engineered plastic material, such as PEEK or PPA.
(29) As illustrated in FIGS. **6** and **7**, the bobbin **130** may have a structure including insertion holes **131** provided inside the bobbin **130** to be open through both sides. The protrusions **112** of the stator core **110**, which have passed through the through-holes **122** of the support structure **120**, are inserted into the insertion holes **131** and the coil **132** is wound on the outer surface of the bobbin **130**.
(30) The bobbin **130** is provided with a plurality of sections matching the plurality of protrusions **112** of the stator core **110** in a one-to-one relationship.
(31) Thus, the protrusions **112** of the stator cores **110**, which have passed through the through-holes **122**, are inserted into the half positions of the insertion holes **131**, respectively, from one side and the other side of the bobbin **130**. The protrusions **112** are thus fastened to the insertion holes **131**. Consequently, as illustrated in FIGS. **6** and **7**, the support structures **120** and the plate portions **111** of the stator cores **110** are in tight contact with one side portion and the other side portion of the bobbin **130**.
(32) The housing **140** may be provided as a structure mounted on the inner circumferential portions and the outer circumferential portions of the support structures **120** so as to seal the bobbin **130**, into which the protrusions **112** of the stator cores **110** are inserted, and the coil **132** wound on the outer circumferential surface of the bobbin **130**.
(33) In this regard, as illustrated in FIGS. **7** and **8**, the housing **140** may include an inner housing **141** joined to the inner diameter surfaces of the inner rings **123** of the support structures **120** by laser welding or thermal fusion. The housing **140** may further include an outer housing **142** joined to the outer diameter surfaces of the outer rings **125** of the support structures **120** by laser welding or thermal fusion.
(34) In this manner, the pair of stator cores **110** are coupled to the stator core-fixing support structures **120** and the protrusions **112** of each of the stator cores **110** are inserted into and fastened to the bobbin **130** on which the coil **132** is wound. Then the housing **140** including the inner housing **141** and the outer housing **142** is fitted to the resultant structure. Consequently, as illustrated in FIG. **8**, the assembly of the stator **100**, according to the present disclosure, may be completed.

(35) In addition, as illustrated in FIG. **8**, the stator **100** according to the present disclosure has an outer appearance by which the outer surfaces of the plate portions **111** of the stator cores **110**, the outer surfaces of the inner rings **123** and the outer rings **125** of the support structures **120**, the inner diameter surface of the inner housing **141**, and the outer diameter of the outer housing **142** are exposed externally. In addition, the bobbin **130**, into which the protrusions **112** of the stator cores **110** are inserted, the coils **132** are wound on the outer surface of the bobbin **130**, and the like, are positioned while being sealed inside the stator **100**.
(36) Particularly, by molding minute (e.g., small, tiny, or the like) gaps of the stator cores **110** and the support structures **120** with an airtight and watertight molding material, the bobbin **130** may be completely sealed.
(37) As illustrated in FIG. **9**, rotors **200** are disposed on both sides of the stator **100** fabricated as described herein at predetermined gap distances from the stator **100**. As illustrated in FIG. **1**, each of the rotors **200** has a structure including a rotor core **201** and a plurality of permanent magnets **202** attached to the inner surface of the rotor core **201**, equidistantly in the circumferential direction.
(38) The performance of the motor may be determined depending on the gap distances between the stator cores **110** of the stator **100** and the permanent magnets **202** of the rotors **200**.
(39) For example, as the gap distances increase between the stator cores **110** of the stator **100** and the permanent magnets **202** of the rotors **200**, there may be degradations in performance of the motor. Thus, increasing the gap distances decreases the average torque (Torque_avg) of the motor, the maximum value (EMF1_pk) of the fundamental wave component of the counter-electromotive force, the maximum value (EMF_pk) of the counter-electromotive force, or the like, as illustrated in FIG. **11**. In contrast, as the gap distances decrease between the stator cores **110** of the stator **100** and the permanent magnets **202** of the rotors **200**, there may be improvements in performance of the motor. Thus, decreasing the gap distances increases the average torque (Torque_avg) of the motor, the maximum value (EMF1_pk) of the fundamental wave component of the counter-electromotive force, the maximum value (EMF_pk) of the counter-electromotive force, or the like, as illustrated in FIG. **11**.
(40) Referring to the left part of FIG. **10** in the accompanying drawings, when a separate cover **150** connected to the support structure is positioned on the outer surface of each of the stator cores **110** or when the support structure is stacked on the outer surface of each of the stator cores **110**, the gap distances between the stator cores **110** and the permanent magnets **202** of the rotors **200** may be increased by the thickness of the cover **150** or the like. Thus, there may be degradations in performance of the motor, such as a decrease in the average torque (Torque_avg) of the motor, the maximum value (EMF1_pk) of the fundamental wave component of the counter-electromotive force, the maximum value (EMF_pk) of the counter-electromotive force, or the like.
(41) In contrast, referring to the right part of FIG. **10** in the accompanying drawings, the plate portion **111** of each of the stator cores **110** according to the present disclosure is inserted into and seated on the recess portion **121** of the corresponding support structure **120** so that the outer surface of the plate portion **111** of the stator core **110** is coplanar with the outer surface of the support structure **120**. Thus, the gap distances between the stator cores **110** and the permanent magnets **202** of the rotors **200** may be minimized, thereby improving the performance of the motor. For example, the average torque (Torque_avg) of the motor, the maximum value (EMF1_pk) of the fundamental wave component of the counter-electromotive force, the maximum value (EMF_pk) of the counter-electromotive force, or the like may be increased.
(42) When the stator core-fixing support structures are formed of aluminum (Al), a non-magnetic material, heat dissipation performance may be obtained due to high thermal conductivity of Al. However, since Al is a conductor, an eddy current loss caused by magnetic field changes may be generated.
(43) Thus, the stator core-fixing support structures **120** according to the present disclosure may be

formed of a non-magnetic engineered plastic material, thereby preventing an electromagnetic loss, such as an eddy current loss.
(44) However, since the stator core-fixing support structures **120** are formed of a non-magnetic engineered plastic material having a low thermal conductivity, cooling performance for the coil **132**, which substantially serves as a heat dissipation component, may degrade.
(45) In order to prevent such a degradation in cooling performance, the bobbin **130** and coil **132** sealed inside the housing **140** and the stator cores **110** may be directly cooled by being immersed in cooling oil.
(46) In this regard, the outer housing **142** is provided with an inlet (e.g., cooling oil inlet) **143** and an outlet (e.g., cooling oil outlet) **144** for the cooling oil. The inlet **143** and the outlet **144** are connected to a cooling device (e.g., a reservoir, an electric pump, or the like) in order to cool the coil **132** wound on the bobbin **130** by immersing the coil **132** in the cooling oil.
(47) Thus, the coil may be directly cooled using the cooling oil by cyclically repeating a cooling process. The cooling may include injecting the cooling oil through the cooling oil inlet **143** formed in the outer housing **142**, cooling the coil **132** wound on the bobbin **130** by immersing the coil **132** in the cooling oil, discharging the cooling oil through the cooling oil outlet **144** formed in the outer housing **142**, and the like. Consequently, the cooling performance for the coil may be improved.
(48) FIG. **12** in the accompanying drawings is a perspective view illustrating stator cores of a stator according to another embodiment of the present disclosure.
(49) As illustrated in FIG. **12**, stator cores **110** according to another embodiment of the present disclosure are characterized by being divided into a plurality of pieces, differently from the integral structure of the stator cores **110** according to the former embodiment.
(50) In this regard, as illustrated in FIG. **12**, each of the stator cores **110** according to another embodiment of the present disclosure may be fabricated to include a structure obtained by stacking and joining a plurality of divided steel pieces **115**, in each of which a protrusion **112** and a plate portion **111** are integrated. The protrusions **112** are configured to extend through the through-holes **122** of a corresponding one of the support structures **120** to be inserted into the insertion holes **131** of the bobbin **130**. The plate portions **111** are configured to be inserted into and seated on the recess portion **121** of a corresponding one of the support structures **120**.
(51) In addition, the protrusion **112** of each of the divided steel pieces **115** of one of the stator cores **110** according to another embodiment has a coupling recess **116** on the distal end thereof. In contrast, the protrusion **112** of each of the divided steel pieces **115** of the other of the stator cores **110** according to another embodiment has a coupling protrusion **117** on the distal end thereof.
(52) Thus, the plate portions **111** of the divided steel pieces **115** are inserted into and seated on the recess portions **121** of the support structures **120** at the same time that the protrusions **112** of the divided steel pieces **115** are inserted into and extend through the through-holes **122** of the support structures **120**. Thus, the outer surfaces of the plate portions **111** of the divided steel pieces **115** and the outer surfaces of the inner ring **123** and the outer ring **125** of the support structures **120** may be disposed coplanar.
(53) Afterward, the protrusions **112** of the divided steel pieces **115**, which have passed through the through-holes **122** of the support structures **120**, may be inserted into and fastened to the insertion holes **131**, respectively, from one side and the other side of the bobbin **130**.
(54) Thus, the protrusions **112** each have the coupling recess **116** of the divided steel pieces **115** inserted into the insertion holes **131** of the bobbin **130**, respectively, from one side of the bobbin **130**, and the protrusions **112** each have the coupling protrusion **117** of the divided steel pieces **115** inserted into the insertion holes **131** of the bobbin **130**, respectively, from the other side of the bobbin **130**. As a result, the coupling protrusions **117** may be inserted into and fastened to the coupling recesses **116**, respectively, as illustrated in FIG. **14**. In this manner, the divided steel pieces **115** facing each other may be coupled to each other, thereby preventing the divided steel pieces **115** from moving.

(55) Accordingly, the support structures **120** and the plate portions **111** of the divided steel pieces **115** of the stator cores **110** are in tight contact with one side and the other side of the bobbin **130**. The plate portions **111** of the divided steel pieces **115** are inserted into and seated on the recess portions **121** of the support structures **120** so as to be coplanar with the outer surfaces of the support structures **120**. In the same manner, the gap distances between the plate portions **111** of the stator cores **110** and the permanent magnets **202** of the rotors **200** may be minimized.
(56) The various embodiments of the present disclosure have been described herein in detail as set forth above, but the scope of the present disclosure is not limited to the described embodiments. It should be apparent to those having ordinary skill in the art that various modifications and improvements made without departing from the technical spirit of the present disclosure defined in the appended claims are also included in the scope of the present disclosure.

Claims

1. A motor stator comprising: a pair of stator cores, each stator core of the pair of stator cores having a structure including a plate portion and a plurality of protrusions provided on one surface of the plate portion; a pair of support structures for fixing the plurality of stator cores, each support structure of the pair of support structures including a plurality of through-holes into which the plurality of protrusions is inserted and a recess portion into which the plate portion is inserted and seated; a bobbin having a plurality of insertion holes formed therein into which each of the plurality of protrusions, having respectively passed through each of the plurality of through-holes of the pair of support structures, is inserted with a coil wound on an outer surface thereof; and a housing structure including an inner housing mounted on inner circumferences of the pair of support structures and an outer housing mounted on outer circumferences of the pair of support structures, wherein the housing includes an inner housing joined to inner diameter surfaces of the pair of support structures and an outer housing joined to outer diameter surfaces of the pair of support structures in order to seal the bobbin on which the coil is wound, wherein the inner housing is joined by laser welding or thermal fusion to the inner diameter surfaces of the pair of support structures, and wherein the outer housing is joined by laser welding or thermal fusion to the outer diameter surfaces of the pair of support structures.
2. The motor stator of claim 1, wherein each of the stator cores has a cylindrical structure with a steel sheet, having a predetermined width, wound a plurality of times around the cylindrical structure and a plurality of cut recess portions on one surface of the cylindrical structure such that the cut recess portions are equidistantly spaced apart from each other in a circumferential direction, so that bottom surfaces of the cut recess portions form the plate portion and the plurality of protrusions are formed to alternate with the cut recess portions.
3. The motor stator of claim 1, wherein each of the support structures is configured such that an inner ring and an outer ring are integrally connected by a plurality of connecting bars equidistantly disposed in a circumferential direction.
4. The motor stator of claim 3, wherein the pair of support structures is fabricated from an engineered plastic material and insert injection molded to be joined to the pair of stator cores, respectively.
5. The motor stator of claim 4, wherein the engineered plastic material is a polyether ether ketone (PEEK) or a polyphthalamide (PPA).
6. The motor stator of claim 3, wherein the through-holes of the support structures are formed to alternate with the connecting bars of the support structures, wherein the recess portion is provided on outer surface portions of the connecting bars connecting the inner ring and the outer ring, and wherein the plate portion of a corresponding one of the pair of stator cores is inserted into and seated on the recess portion.
7. The motor stator of claim 6, wherein the inner ring and the outer ring have a same thickness, a

thickness of each of the connecting bars is lower than the thickness of either the inner ring or the outer ring, and the recess portion is formed on the outer surface portions of the connecting bars connecting an outer diameter surface of the inner ring and an inner diameter surface of the outer ring.

8. The motor stator of claim 6, wherein the depth of each of the recess portions of the support structures is the same as a thickness of each of the plate portions of the stator cores.

9. The motor stator of claim 6, wherein, when the plate portion of each of the stator cores is inserted into and seated on the recess portion of a corresponding one of the support structures, an outer surface of the plate portion of the stator core and the outer surfaces of the inner ring and the outer ring of the support structure are disposed coplanar.

10. The motor stator of claim 1, wherein the outer housing comprises a cooling oil inlet and a cooling oil outlet configured to cool the coil wound on the bobbin by immersing the coil in cooling oil.

11. The motor stator of claim 1, wherein each stator core includes a plurality of divided steel pieces, stacked and joined, and in each of which a protrusion and a plate portion are integrated, wherein the protrusions are configured to extend through the through-holes of a corresponding one of the pair of support structures and be inserted into the insertion holes of the bobbin, and wherein the plate portions are configured to be inserted into and seated on the recess portion of a corresponding one of the pair of support structures.

12. The motor stator of claim 11, wherein the protrusion of each of the divided steel pieces of the stator cores has a coupling recess or a coupling protrusion on a distal end thereof.

13. The motor stator of claim 12, wherein, when the protrusions each having the coupling recess of the divided steel pieces are inserted into the insertion holes of the bobbin, respectively, from one side of the bobbin and the protrusions each having the coupling protrusion of the divided steel pieces are inserted into the insertion holes of the bobbin, respectively, from the other side of the bobbin, the coupling protrusions are inserted into and fastened to the coupling recesses, respectively.